

Lab 4

Building a Modified kernel

Approach: Start by getting the minix machine back up and running with the image that is provided. Once in root, make another image so I could change the kernel to do what I wanted. Once that was recompiled and running on the new image, find where the kernel is idle as in which c file has the code. After I found it, I added the print statement to the c file and recompiled the image again. Run the minix machine on the new kernel and it should have the @ symbol on the input space.

Problems Encountered: I had a couple of problems, but I don't think that they were really my fault. First, for some reason, I couldn't use the qemu on the CSLs properly so I had to have a flag that would display in the terminal which made the minix machine really slow on loading. It took like 5 minutes. After I got that done, I had some problems with adding the line into the c file because I was using vi instead of vim so that took some time getting used to. Once that was done and the print statement was in the right spot, I wouldn't put the @ symbol on the right spot. I would appear after I put spaces and then backspaced but not when it was idle. Prof. Nico even said that it was in the right spot.

Solutions: To make it easier on myself and to make sure that the minix machine would work, I installed it onto my own computer instead of using the CSLs. This made it so that the qemu window would pop up, it was much faster at loading the machine, and it made the print statement work when it idled.

Lessons Learned: Using stuff on the CSL is good, but maybe if it's gonna be a big assignment later, I should install it onto my own computer so I have less problems. Also having Prof. Nico there to help is wayyy better than suffering in silence.

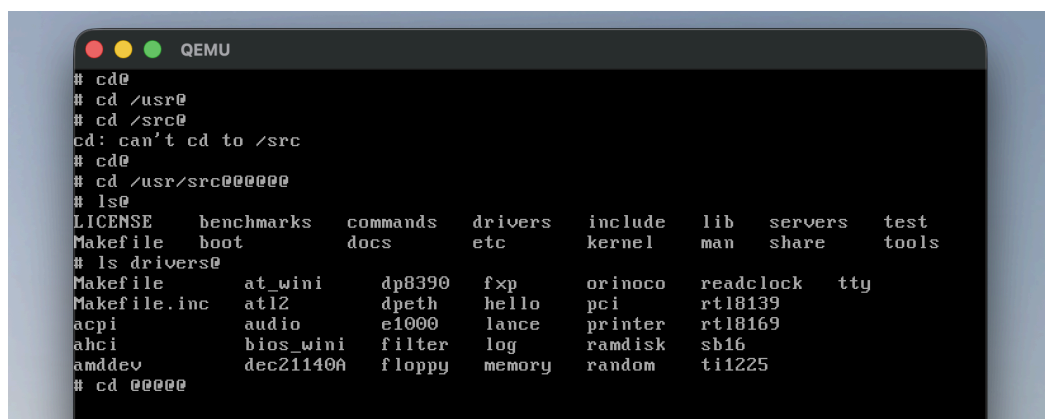
Testing my code

Approach: I changed directories into different places and also typed and then deleted which also led to the @ symbol print. Whenever I am not typing, the system is idle so the @ symbol appears all the time.

Problems: No problems!

Solutions: If I got no problems, I don't need any solutions

Lessons Learned: Messing around with the minix machine is fun and interesting to go through.



```
# cd@
# cd /usr@
# cd /src@
cd: can't cd to /src
# cd@
# cd /usr/src@
# ls@
LICENSE      benchmarks  commands   drivers     include     lib         servers     test
Makefile     boot       docs        etc         kernel      man         share       tools
# ls drivers@
Makefile      at_wini      dp8390      fxp         orinoco      readclock   tty
Makefile.inc  at12         dpeth       hello       pci          rtl8139
acpi          audio        e1000       lance       printer      rtl8169
ahci          bios_wini    filter      log         ramdisk      sb16
amdddev       dec21140a    floppy     memory      random       ti1225
# cd @
```